

The diagram on the left shows a quark-antiquark pair (represented by solid lines with arrows) interacting via a gluon exchange (represented by a wavy line). The incoming quark has momentum $p_1 - k$ and the incoming antiquark has momentum $p_2 - k$. The outgoing quark has momentum $p_1 - k$ and the outgoing antiquark has momentum $p_2 - k$. The gluon exchange is labeled k . The diagram is split by a vertical red line, indicating a soft gluon emission.

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \left(\left(\text{Diagram} \right) \Big|_{\text{soft}} \right) + \mathcal{O} \left(\frac{\Lambda^2}{q_h^2} \right)$$

The diagram inside the large parentheses is a circular loop diagram. The top half is a quark-antiquark pair with momenta k and p_1 . The bottom half is a gluon loop with momenta p_2 and k . A red line points to a red circle with a cross inside the gluon loop, indicating a soft gluon emission.